Fortinet Capstone: Full Device Configuration

About this document

This is the final configuration for every device in the lab, after all fixes found during testing. It uses Cisco IOS CLI for routers and switches, and FortiOS CLI for the FortiGates. Every FortiGate block also names the GUI path, if you prefer the GUI.

Changes from the earlier drafts:

- The IPsec tunnels use `des-md5`, because the lab VM rejected AES. Both sides must match.
- port5 on every FortiGate has `defaultgw disable`, so it does not steal the default route.
- Cisco interfaces that were administratively down after a restart all have `no shutdown`.
- The ISP Gi0/1 addresses are static, so they do not change after a restart.

Paste tips:

- Paste in small blocks. A long paste can cut lines off with `$` characters.
- Run `write memory` on every Cisco device when done. FortiGate saves automatically.
- Replace `<HA-PASSWORD>` and `<STRONG-PASSWORD>` with your own values.

Interface map (matches the topology)

```
HQ-ISP    Gi0/1 -> Net (10.0.137.91)       Gi0/0 -> HQ-Edge Gi0/0 (203.0.113.1/30)
HQ-Edge   Gi0/0 -> HQ-ISP (203.0.113.2/30) Gi0/1 -> HQ-WAN-SW Gi0/0 (10.100.0.1/29)
HQ-WAN-SW Gi0/0 HQ-Edge, Gi0/1 HQ-FG1 port1, Gi0/2 HQ-FG2 port1
HQ-FG1/2  port1 WAN 10.100.0.2/29 (shared) | port2 trunk to HQ-Core Gi0/0, Gi0/1
          port3, port4 HA heartbeat        | port5 management (MGMT cloud)
HQ-Core   Gi0/0 FG1 port2, Gi0/1 FG2 port2, Gi0/2 HQ-User, Gi0/3 HQ-SRV-MGMT
Br1-ISP   Gi0/1 -> Net (10.0.137.181)      Gi0/0 -> Br1-Edge (203.0.113.5/30)
Br1-Edge  Gi0/0 (203.0.113.6/30)           Gi0/1 -> Br1-FG port1 (10.100.1.1/29)
Br1-FG    port1 10.100.1.2/29 | port2 trunk to Br1-Core Gi0/0 | port5 MGMT
Br1-Core  Gi0/0 trunk, Gi0/1 VPC
Br2-ISP   Gi0/1 -> Net (10.0.137.46)       Gi0/0 -> Br2-Edge (203.0.113.9/30)
Br2-Edge  Gi0/0 (203.0.113.10/30)          Gi0/1 -> Br2-FG port1 (10.100.2.1/29)
Br2-FG    port1 10.100.2.2/29 | port2 trunk to Br2-Core Gi0/0 | port5 MGMT
Br2-Core  Gi0/0 trunk, Gi0/1 VPC
RemoteUser e0 -> Net (DHCP)
```

Part 1: Cisco devices

Common lines (every router and switch, first)

```
enable
configure terminal
no ip domain-lookup
line con 0
 logging synchronous
 exec-timeout 0 0
 exit
```

HQ-ISP

Current Gi0/1 is `10.0.137.91`, Br1-ISP is `10.0.137.181`, Br2-ISP is `10.0.137.46`. Run `show ip interface Gi0/1` first to confirm the mask and the gateway `10.0.137.1`. The mask is `/24` in this example.

```
hostname HQ-ISP
interface Gi0/1
 description To Net cloud
 ip address 10.0.137.91 255.255.255.0
 ip nat outside
 no shutdown
interface Gi0/0
 description To HQ-Edge Gi0/0
 ip address 203.0.113.1 255.255.255.252
 ip nat inside
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 10.0.137.1
ip route 10.100.0.0 255.255.255.248 203.0.113.2
ip route 10.100.1.0 255.255.255.248 10.0.137.181
ip route 10.100.2.0 255.255.255.248 10.0.137.46
ip access-list extended NAT-OUT
 deny ip 10.100.0.0 0.0.255.255 10.100.0.0 0.0.255.255
 permit ip 10.100.0.0 0.0.255.255 any
 exit
ip nat inside source list NAT-OUT interface Gi0/1 overload
ip nat inside source static tcp 10.100.0.2 10443 interface Gi0/1 10443
ip nat inside source static tcp 10.100.0.2 8080 interface Gi0/1 8080
end
write memory
```

If you prefer DHCP, replace the Gi0/1 address line with `ip address dhcp`, remove the `ip route 0.0.0.0` line, and update the routes after every restart.

HQ-Edge

```
hostname HQ-Edge
interface Gi0/0
 description To HQ-ISP Gi0/0
 ip address 203.0.113.2 255.255.255.252
 no shutdown
interface Gi0/1
 description To HQ-WAN-SW Gi0/0
 ip address 10.100.0.1 255.255.255.248
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
write memory
```

Switch (HQ WAN switch)

Gi0/0 goes to HQ-Edge, Gi0/1 to HQ-FG1 port1, Gi0/2 to HQ-FG2 port1.

```
hostname HQ-WAN-SW
interface range Gi0/0 - 2
 switchport mode access
 spanning-tree portfast
```

```
 no shutdown
end
write memory
```

HQ-Core (Layer 2 only)

Gi0/0 goes to FG1 port2, Gi0/1 to FG2 port2, Gi0/2 to HQ-User, Gi0/3 to HQ-SRV-MGMT.

```
hostname HQ-Core
no ip routing
vlan 10
 name Users
vlan 20
 name Servers
vlan 30
 name Mgmt
vlan 80
 name DMZ
 exit
spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,80 priority 4096
interface range Gi0/0 - 3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,80
 no shutdown
 exit
interface Vlan30
 ip address 10.10.30.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.10.30.1
end
write memory
```

HQ-User

```
hostname HQ-User
vlan 10
 name Users
vlan 30
 name Mgmt
 exit
interface Gi0/0
 description Trunk to HQ-Core Gi0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,30
 no shutdown
interface range Gi0/1 - 3
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
 exit
interface Vlan30
 ip address 10.10.30.3 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.10.30.1
end
write memory
```

HQ-SRV-MGMT

```
hostname HQ-SRV-MGMT
vlan 20
 name Servers
vlan 30
 name Mgmt
vlan 80
 name DMZ
 exit
```

```
interface Gi0/0
 description Trunk to HQ-Core Gi0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,30,80
 no shutdown
interface Gi0/1
 description HQ server VLAN 20
 switchport mode access
 switchport access vlan 20
 spanning-tree portfast
 no shutdown
interface Gi0/2
 description DMZ web server VLAN 80
 switchport mode access
 switchport access vlan 80
 spanning-tree portfast
 no shutdown
interface Gi0/3
 description Syslog VLAN 30
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown
 exit
interface Vlan30
 ip address 10.10.30.4 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.10.30.1
end
write memory
```

Br1-ISP

Static Gi0/1 `10.0.137.181`. For DHCP, use `ip address dhcp` and drop the default route.

```
hostname Br1-ISP
interface Gi0/1
 description To Net cloud
 ip address 10.0.137.181 255.255.255.0
 ip nat outside
 no shutdown
interface Gi0/0
 description To Br1-Edge Gi0/0
 ip address 203.0.113.5 255.255.255.252
 ip nat inside
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 10.0.137.1
ip route 10.100.1.0 255.255.255.248 203.0.113.6
ip route 10.100.0.0 255.255.255.248 10.0.137.91
ip route 10.100.2.0 255.255.255.248 10.0.137.46
ip access-list extended NAT-OUT
 deny ip 10.100.0.0 0.0.255.255 10.100.0.0 0.0.255.255
 permit ip 10.100.0.0 0.0.255.255 any
 exit
ip nat inside source list NAT-OUT interface Gi0/1 overload
end
write memory
```

Br1-Edge

```
hostname Br1-Edge
interface Gi0/0
 description To Br1-ISP Gi0/0
 ip address 203.0.113.6 255.255.255.252
 no shutdown
interface Gi0/1
 description To Br1-FG port1
 ip address 10.100.1.1 255.255.255.248
 no shutdown
 exit
```

```
ip route 0.0.0.0 0.0.0.0 203.0.113.5
end
write memory
```

Br1-Core

VLANs 40 and 50 match the Br1-FG interfaces.

```
hostname Br1-Core
no ip routing
vlan 10
 name Users
vlan 20
 name Servers
vlan 30
 name Mgmt
vlan 40
 name Guest
vlan 50
 name IoT
 exit
spanning-tree mode rapid-pvst
interface Gi0/0
 description Trunk to Br1-FG port2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50
 no shutdown
interface Gi0/1
 description VPC Users
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
 exit
interface Vlan30
 ip address 10.11.30.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.11.30.1
end
write memory
```

Br2-ISP

```
hostname Br2-ISP
interface Gi0/1
 description To Net cloud
 ip address 10.0.137.46 255.255.255.0
 ip nat outside
 no shutdown
interface Gi0/0
 description To Br2-Edge Gi0/0
 ip address 203.0.113.9 255.255.255.252
 ip nat inside
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 10.0.137.1
ip route 10.100.2.0 255.255.255.248 203.0.113.10
ip route 10.100.0.0 255.255.255.248 10.0.137.91
ip route 10.100.1.0 255.255.255.248 10.0.137.181
ip access-list extended NAT-OUT
 deny ip 10.100.0.0 0.0.255.255 10.100.0.0 0.0.255.255
 permit ip 10.100.0.0 0.0.255.255 any
 exit
ip nat inside source list NAT-OUT interface Gi0/1 overload
end
write memory
```

Br2-Edge

```
hostname Br2-Edge
interface Gi0/0
```

```
 description To Br2-ISP Gi0/0
 ip address 203.0.113.10 255.255.255.252
 no shutdown
interface Gi0/1
 description To Br2-FG port1
 ip address 10.100.2.1 255.255.255.248
 no shutdown
 exit
ip route 0.0.0.0 0.0.0.0 203.0.113.9
end
write memory
```

Br2-Core

```
hostname Br2-Core
no ip routing
vlan 10
 name Users
vlan 20
 name Servers
vlan 30
 name Mgmt
 exit
spanning-tree mode rapid-pvst
interface Gi0/0
 description Trunk to Br2-FG port2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
interface Gi0/1
 description VPC Users
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
 exit
interface Vlan30
 ip address 10.12.30.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.12.30.1
end
write memory
```

If `switchport trunk encapsulation dot1q` is rejected on any switch, delete that one line.

VPCs and hosts

```
HQ PC (behind HQ-User):   ip dhcp
Br1 VPC:                  ip dhcp
Br2 VPC:                  ip dhcp
HQ server (VLAN 20):      ip 10.10.20.10/24 10.10.20.1
DMZ web server (VLAN 80): ip 10.10.80.10/24 10.10.80.1
```

The DMZ web server must run a real web service on port 80. RemoteUser must be a real OS (Windows or Ubuntu), not a VPCS.

Part 2: HQ FortiGates

Interface names below are `HQ-USERS`, `HQ-SERVERS`, `HQ-MGMT`, `HQ-DMZ`. If you used other names, replace them everywhere.

2A. HQ-FG1 and HQ-FG2: HA (GUI: System > HA)

HQ-FG1:

```
config system ha
 set group-name "HQ-HA"
 set mode a-p
 set password <HA-PASSWORD>
 set hbdev "port3" 100 "port4" 50
 set session-pickup enable
 set priority 200
 set monitor "port1" "port2"
 set hostname "HQ-FG1"
 set ha-mgmt-status enable
 config ha-mgmt-interfaces
  edit 1
   set interface "port5"
   set gateway 192.168.174.2
  next
 end
end
config system interface
 edit "port5"
  set mode static
  set ip 192.168.174.166 255.255.255.0
  set allowaccess https ping ssh
 next
end
```

HQ-FG2 (same, but different priority, hostname and port5 IP):

```
config system ha
 set group-name "HQ-HA"
 set mode a-p
 set password <HA-PASSWORD>
 set hbdev "port3" 100 "port4" 50
 set session-pickup enable
 set priority 100
 set monitor "port1" "port2"
 set hostname "HQ-FG2"
 set ha-mgmt-status enable
 config ha-mgmt-interfaces
  edit 1
   set interface "port5"
   set gateway 192.168.174.2
  next
 end
end
config system interface
 edit "port5"
  set mode static
  set ip 192.168.174.167 255.255.255.0
  set allowaccess https ping ssh
 next
end
```

- The password must be the same on both units.
- The gateway `192.168.174.2` was a guess. Use your real VMware gateway.
- Start FG1 first so it wins the election. Check both units with `get system ha status`.
- **Everything below is configured on FG1 only.** It syncs to FG2.

2B. HQ-FG1: interfaces (GUI: Network > Interfaces)

```
config system interface
 edit "port1"
```

```
    set mode static
    set ip 10.100.0.2 255.255.255.248
    set allowaccess ping
    set role wan
  next
  edit "port2"
    set mode static
    set role lan
  next
  edit "HQ-USERS"
    set vdom "root"
    set interface "port2"
    set vlanid 10
    set ip 10.10.10.1 255.255.255.0
    set allowaccess ping
    set role lan
  next
  edit "HQ-SERVERS"
    set vdom "root"
    set interface "port2"
    set vlanid 20
    set ip 10.10.20.1 255.255.255.0
    set allowaccess ping
    set role lan
  next
  edit "HQ-MGMT"
    set vdom "root"
    set interface "port2"
    set vlanid 30
    set ip 10.10.30.1 255.255.255.0
    set allowaccess ping https ssh
    set role lan
  next
  edit "HQ-DMZ"
    set vdom "root"
    set interface "port2"
    set vlanid 80
    set ip 10.10.80.1 255.255.255.0
    set allowaccess ping
    set role dmz
  next
end
```

2C. DNS and DHCP (GUI: Network > DNS, Network > DHCP Server)

```
config system dns
  set primary 8.8.8.8
  set secondary 8.8.4.4
end
config system dhcp server
  edit 1
    set interface "HQ-USERS"
    set default-gateway 10.10.10.1
    set netmask 255.255.255.0
    set dns-service default
    config ip-range
      edit 1
        set start-ip 10.10.10.50
        set end-ip 10.10.10.200
      next
    end
  next
end
```

Use `edit 2` if `edit 1` already exists. Check with `show system dhcp server`.

2D. Address objects (GUI: Policy & Objects > Addresses)

```
config firewall address
  edit "HQ-Users"
    set subnet 10.10.10.0 255.255.255.0
  next
```

```
 edit "HQ-Servers"
  set subnet 10.10.20.0 255.255.255.0
 next
 edit "HQ-Mgmt"
  set subnet 10.10.30.0 255.255.255.0
 next
 edit "HQ-DMZ"
  set subnet 10.10.80.0 255.255.255.0
 next
 edit "HQ-LAN"
  set subnet 10.10.0.0 255.255.0.0
 next
 edit "Branch1-LAN"
  set subnet 10.11.0.0 255.255.0.0
 next
 edit "Branch2-LAN"
  set subnet 10.12.0.0 255.255.0.0
 next
 edit "VPN-Pool"
  set type iprange
  set start-ip 10.20.0.10
  set end-ip 10.20.0.50
 next
end
```

2E. Security profiles (GUI: Security Profiles)

```
config antivirus profile
 edit "LAB-AV"
  config http
   set av-scan block
  end
  config ftp
   set av-scan block
  end
  config smtp
   set av-scan block
  end
  config pop3
   set av-scan block
  end
  config imap
   set av-scan block
  end
 next
end
config webfilter urlfilter
 edit 1
  set name "LAB-URLS"
  config entries
   edit 1
    set url "*.blocked-test.com"
    set type wildcard
    set action block
   next
  end
 next
end
config webfilter profile
 edit "LAB-WEB"
  config web
   set urlfilter-table 1
  end
 next
end
config application list
 edit "LAB-APP"
  config entries
   edit 1
    set category 2
    set action block
   next
   edit 2
```

```
    set category 6
    set action block
   next
  end
 next
end
```

- **IPS:** use the built-in sensor `default`. No config is needed.
- **SSL inspection:** use the built-in `certificate-inspection`.
- **Application Control:** category 2 is P2P and 6 is Proxy. If your build differs, type `set category ?` to check.
- **Web filter categories:** add Malicious Websites, Phishing, Proxy Avoidance and Adult in the GUI (Security Profiles > Web Filter > LAB-WEB). They need FortiGuard access. The static URL filter works without it.

2F. IPsec tunnels (GUI: VPN > IPsec Tunnels > Custom)

HQ-to-Br2 is already configured. HQ-to-Br1 uses the same settings. The PSK `123456` is a lab key only. Use a stronger key in your report if you can.

```
config vpn ipsec phase1-interface
 edit "to-B1"
  set interface "port1"
  set ike-version 2
  set peertype any
  set net-device disable
  set proposal des-md5
  set remote-gw 10.100.1.2
  set psk 123456
 next
 edit "to-B2"
  set interface "port1"
  set ike-version 2
  set peertype any
  set net-device disable
  set proposal des-md5
  set remote-gw 10.100.2.2
  set psk 123456
 next
end
config vpn ipsec phase2-interface
 edit "to-B1"
  set phase1name "to-B1"
  set proposal des-md5 des-sha1
  set auto-negotiate enable
 next
 edit "to-B2"
  set phase1name "to-B2"
  set proposal des-md5 des-sha1
  set auto-negotiate enable
 next
end
```

Phase 2 selectors stay at the default 0.0.0.0/0, which lets Branch 1 reach Branch 2 through HQ.

2G. Static routes (GUI: Network > Static Routes)

```
config router static
 edit 1
  set gateway 10.100.0.1
  set device "port1"
 next
 edit 2
  set dst 10.11.0.0 255.255.0.0
  set device "to-B1"
 next
 edit 3
  set dst 10.12.0.0 255.255.0.0
  set device "to-B2"
 next
end
```

Edit numbers must be free. Check with `show router static`.

2H. Virtual IP (GUI: Policy & Objects > Virtual IPs)

```
config firewall vip
 edit "Web-VIP"
  set extintf "port1"
  set extip 10.100.0.2
  set mappedip "10.10.80.10"
  set portforward enable
  set protocol tcp
  set extport 8080
  set mappedport 80
 next
end
```

2I. SSL-VPN for RemoteUser (GUI: VPN > SSL-VPN Settings)

This works on FortiOS 7.2 and 7.4. On 7.6.3 and later, SSL-VPN is removed, so use an IPsec dial-up tunnel with FortiClient instead.

```
config user local
 edit "remote1"
  set type password
  set passwd <STRONG-PASSWORD>
 next
end
config user group
 edit "SSL-VPN-Users"
  set member "remote1"
 next
end
config vpn ssl settings
 set servercert "Fortinet_Factory"
 set tunnel-ip-pools "VPN-Pool"
 set dns-server1 8.8.8.8
 set port 10443
 set source-interface "port1"
 set source-address "all"
 set default-portal "full-access"
 config authentication-rule
  edit 1
   set groups "SSL-VPN-Users"
   set portal "full-access"
  next
 end
end
```

2J. Firewall policies (GUI: Policy & Objects > Firewall Policy)

Security profiles used in the policies: `ssl-ssh-profile "certificate-inspection"`, `av-profile "LAB-AV"`, `webfilter-profile "LAB-WEB"`, `ips-sensor "default"`, `application-list "LAB-APP"`.

```
config firewall policy
 edit 1
  set name "Users-to-Servers"
  set srcintf "HQ-USERS"
  set dstintf "HQ-SERVERS"
  set srcaddr "HQ-Users"
  set dstaddr "HQ-Servers"
  set action accept
  set schedule "always"
  set service "DNS" "HTTP" "HTTPS" "SMB" "PING"
  set utm-status enable
  set ssl-ssh-profile "certificate-inspection"
  set ips-sensor "default"
  set application-list "LAB-APP"
  set logtraffic all
 next
 edit 2
  set name "Users-to-DMZ"
  set srcintf "HQ-USERS"
```

```
    set dstintf "HQ-DMZ"
    set srcaddr "HQ-Users"
    set dstaddr "HQ-DMZ"
    set action accept
    set schedule "always"
    set service "HTTP" "HTTPS" "PING"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set ips-sensor "default"
    set logtraffic all
  next
  edit 3
    set name "Mgmt-to-All"
    set srcintf "HQ-MGMT"
    set dstintf "HQ-USERS" "HQ-SERVERS" "HQ-DMZ"
    set srcaddr "HQ-Mgmt"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set logtraffic all
  next
  edit 4
    set name "Users-to-Internet"
    set srcintf "HQ-USERS"
    set dstintf "port1"
    set srcaddr "HQ-Users"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat enable
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 5
    set name "Servers-to-Internet"
    set srcintf "HQ-SERVERS"
    set dstintf "port1"
    set srcaddr "HQ-Servers"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "DNS" "HTTP" "HTTPS" "NTP"
    set nat enable
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 6
    set name "Internet-to-Web"
    set srcintf "port1"
    set dstintf "HQ-DMZ"
    set srcaddr "all"
    set dstaddr "Web-VIP"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set ips-sensor "default"
    set logtraffic all
  next
  edit 7
```

```
    set name "HQ-to-Branches"
    set srcintf "HQ-USERS" "HQ-SERVERS"
    set dstintf "to-B1" "to-B2"
    set srcaddr "HQ-Users" "HQ-Servers"
    set dstaddr "Branch1-LAN" "Branch2-LAN"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 8
    set name "Branches-to-HQ"
    set srcintf "to-B1" "to-B2"
    set dstintf "HQ-USERS" "HQ-SERVERS"
    set srcaddr "Branch1-LAN" "Branch2-LAN"
    set dstaddr "HQ-Users" "HQ-Servers"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 9
    set name "Branch1-to-Branch2"
    set srcintf "to-B1"
    set dstintf "to-B2"
    set srcaddr "Branch1-LAN"
    set dstaddr "Branch2-LAN"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 10
    set name "Branch2-to-Branch1"
    set srcintf "to-B2"
    set dstintf "to-B1"
    set srcaddr "Branch2-LAN"
    set dstaddr "Branch1-LAN"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 11
    set name "SSLVPN-to-Servers"
    set srcintf "ssl.root"
    set dstintf "HQ-SERVERS"
    set srcaddr "VPN-Pool"
    set dstaddr "HQ-Servers"
```

```
    set groups "SSL-VPN-Users"
    set action accept
    set schedule "always"
    set service "HTTP" "HTTPS" "SMB" "PING"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set ips-sensor "default"
    set logtraffic all
  next
  edit 12
    set name "Deny-All-Log"
    set srcintf "any"
    set dstintf "any"
    set srcaddr "all"
    set dstaddr "all"
    set action deny
    set schedule "always"
    set service "ALL"
    set logtraffic all
  next
end
```

If a policy ID already exists, FortiOS edits it. Check the order with `show firewall policy`. The deny-all must stay last.

Part 3: Br1-FG

3A. Hostname, interfaces, DNS and DHCP

```
config system global
 set hostname "Br1-FG"
end
config system interface
 edit "port1"
  set mode static
  set ip 10.100.1.2 255.255.255.248
  set allowaccess ping
  set role wan
 next
 edit "port2"
  set mode static
  set role lan
 next
 edit "port5"
  set mode dhcp
  set defaultgw disable
  set allowaccess https ping
 next
 edit "B1-USERS"
  set vdom "root"
  set interface "port2"
  set vlanid 10
  set ip 10.11.10.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
 edit "B1-SERVERS"
  set vdom "root"
  set interface "port2"
  set vlanid 20
  set ip 10.11.20.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
 edit "B1-MGMT"
  set vdom "root"
  set interface "port2"
  set vlanid 30
  set ip 10.11.30.1 255.255.255.0
  set allowaccess ping https ssh
  set role lan
 next
 edit "B1-GUEST"
  set vdom "root"
  set interface "port2"
  set vlanid 40
  set ip 10.11.40.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
 edit "B1-IOT"
  set vdom "root"
  set interface "port2"
  set vlanid 50
  set ip 10.11.50.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
end
config system dns
 set primary 8.8.8.8
 set secondary 8.8.4.4
end
config system dhcp server
 edit 1
  set interface "B1-USERS"
  set default-gateway 10.11.10.1
```

```
  set netmask 255.255.255.0
  set dns-service default
  config ip-range
   edit 1
    set start-ip 10.11.10.50
    set end-ip 10.11.10.200
   next
  end
 next
end
```

3B. Address objects and security profiles

```
config firewall address
 edit "B1-Users"
  set subnet 10.11.10.0 255.255.255.0
 next
 edit "B1-Servers"
  set subnet 10.11.20.0 255.255.255.0
 next
 edit "B1-LAN"
  set subnet 10.11.0.0 255.255.0.0
 next
 edit "HQ-LAN"
  set subnet 10.10.0.0 255.255.0.0
 next
 edit "Branch2-LAN"
  set subnet 10.12.0.0 255.255.0.0
 next
end
```

Create `LAB-AV`, `LAB-URLS`, `LAB-WEB` and `LAB-APP` with the same commands as in section 2E.

3C. IPsec tunnel (the PSK must match HQ to-B1)

```
config vpn ipsec phase1-interface
 edit "Br1-to-HQ"
  set interface "port1"
  set ike-version 2
  set peertype any
  set net-device disable
  set proposal des-md5
  set remote-gw 10.100.0.2
  set psk 123456
 next
end
config vpn ipsec phase2-interface
 edit "Br1-to-HQ"
  set phase1name "Br1-to-HQ"
  set proposal des-md5 des-sha1
  set auto-negotiate enable
 next
end
```

3D. Static routes

```
config router static
 edit 1
  set gateway 10.100.1.1
  set device "port1"
 next
 edit 2
  set dst 10.10.0.0 255.255.0.0
  set device "Br1-to-HQ"
 next
 edit 3
  set dst 10.12.0.0 255.255.0.0
  set device "Br1-to-HQ"
 next
end
```

3E. Firewall policies

```
config firewall policy
 edit 1
  set name "Users-to-Servers"
  set srcintf "B1-USERS"
  set dstintf "B1-SERVERS"
  set srcaddr "B1-Users"
  set dstaddr "B1-Servers"
  set action accept
  set schedule "always"
  set service "HTTP" "HTTPS" "SMB" "PING"
  set utm-status enable
  set ssl-ssh-profile "certificate-inspection"
  set ips-sensor "default"
  set logtraffic all
 next
 edit 2
  set name "Users-to-Internet"
  set srcintf "B1-USERS"
  set dstintf "port1"
  set srcaddr "B1-Users"
  set dstaddr "all"
  set action accept
  set schedule "always"
  set service "ALL"
  set nat enable
  set utm-status enable
  set ssl-ssh-profile "certificate-inspection"
  set av-profile "LAB-AV"
  set webfilter-profile "LAB-WEB"
  set ips-sensor "default"
  set application-list "LAB-APP"
  set logtraffic all
 next
 edit 3
  set name "Servers-to-Internet"
  set srcintf "B1-SERVERS"
  set dstintf "port1"
  set srcaddr "B1-Servers"
  set dstaddr "all"
  set action accept
  set schedule "always"
  set service "DNS" "HTTP" "HTTPS" "NTP"
  set nat enable
  set utm-status enable
  set ssl-ssh-profile "certificate-inspection"
  set av-profile "LAB-AV"
  set webfilter-profile "LAB-WEB"
  set ips-sensor "default"
  set application-list "LAB-APP"
  set logtraffic all
 next
 edit 4
  set name "LAN-to-VPN"
  set srcintf "B1-USERS" "B1-SERVERS"
  set dstintf "Br1-to-HQ"
  set srcaddr "B1-Users" "B1-Servers"
  set dstaddr "HQ-LAN" "Branch2-LAN"
  set action accept
  set schedule "always"
  set service "ALL"
  set utm-status enable
  set ssl-ssh-profile "certificate-inspection"
  set av-profile "LAB-AV"
  set webfilter-profile "LAB-WEB"
  set ips-sensor "default"
  set application-list "LAB-APP"
  set logtraffic all
 next
 edit 5
  set name "VPN-to-LAN"
  set srcintf "Br1-to-HQ"
```

```
    set dstintf "B1-USERS" "B1-SERVERS"
    set srcaddr "HQ-LAN" "Branch2-LAN"
    set dstaddr "B1-Users" "B1-Servers"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
  next
  edit 6
    set name "Deny-All-Log"
    set srcintf "any"
    set dstintf "any"
    set srcaddr "all"
    set dstaddr "all"
    set action deny
    set schedule "always"
    set service "ALL"
    set logtraffic all
  next
end
```

Guest (VLAN 40) and IoT (VLAN 50) have no allow policy, so they are denied by default.

Part 4: Br2-FG

4A. Hostname, interfaces, DNS and DHCP

```
config system global
 set hostname "Br2-FG"
end
config system interface
 edit "port1"
  set mode static
  set ip 10.100.2.2 255.255.255.248
  set allowaccess ping
  set role wan
 next
 edit "port2"
  set mode static
  set role lan
 next
 edit "port5"
  set mode dhcp
  set defaultgw disable
  set allowaccess https ping
 next
 edit "B2-USERS"
  set vdom "root"
  set interface "port2"
  set vlanid 10
  set ip 10.12.10.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
 edit "B2-SERVERS"
  set vdom "root"
  set interface "port2"
  set vlanid 20
  set ip 10.12.20.1 255.255.255.0
  set allowaccess ping
  set role lan
 next
 edit "B2-MGMT"
  set vdom "root"
  set interface "port2"
  set vlanid 30
  set ip 10.12.30.1 255.255.255.0
  set allowaccess ping https ssh
  set role lan
 next
end
config system dns
 set primary 8.8.8.8
 set secondary 8.8.4.4
end
config system dhcp server
 edit 1
  set interface "B2-USERS"
  set default-gateway 10.12.10.1
  set netmask 255.255.255.0
  set dns-service default
  config ip-range
   edit 1
    set start-ip 10.12.10.50
    set end-ip 10.12.10.200
   next
  end
 next
end
```

If `set defaultgw disable` gives a parse error, port5 is still static. Run `set mode dhcp` first. That is what happened on your unit.

4B. Address objects and security profiles

```
config firewall address
 edit "B2-Users"
  set subnet 10.12.10.0 255.255.255.0
 next
 edit "B2-Servers"
  set subnet 10.12.20.0 255.255.255.0
 next
 edit "B2-LAN"
  set subnet 10.12.0.0 255.255.0.0
 next
 edit "HQ-LAN"
  set subnet 10.10.0.0 255.255.0.0
 next
 edit "Branch1-LAN"
  set subnet 10.11.0.0 255.255.0.0
 next
end
```

Create `LAB-AV`, `LAB-URLS`, `LAB-WEB` and `LAB-APP` with the same commands as in section 2E.

4C. IPsec tunnel (matches HQ to-B2)

```
config vpn ipsec phase1-interface
 edit "Br2-to-HQ"
  set interface "port1"
  set ike-version 2
  set peertype any
  set net-device disable
  set proposal des-md5
  set remote-gw 10.100.0.2
  set psk 123456
 next
end
config vpn ipsec phase2-interface
 edit "Br2-to-HQ"
  set phase1name "Br2-to-HQ"
  set proposal des-md5 des-sha1
  set auto-negotiate enable
 next
end
```

4D. Static routes

```
config router static
 edit 1
  set gateway 10.100.2.1
  set device "port1"
 next
 edit 2
  set dst 10.10.0.0 255.255.0.0
  set device "Br2-to-HQ"
 next
 edit 3
  set dst 10.11.0.0 255.255.0.0
  set device "Br2-to-HQ"
 next
end
```

4E. Firewall policies

```
config firewall policy
 edit 1
  set name "Users-to-Servers"
  set srcintf "B2-USERS"
  set dstintf "B2-SERVERS"
  set srcaddr "B2-Users"
  set dstaddr "B2-Servers"
  set action accept
  set schedule "always"
  set service "HTTP" "HTTPS" "SMB" "PING"
```

```
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set ips-sensor "default"
    set logtraffic all
   next
   edit 2
    set name "Users-to-Internet"
    set srcintf "B2-USERS"
    set dstintf "port1"
    set srcaddr "B2-Users"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "ALL"
    set nat enable
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
   next
   edit 3
    set name "Servers-to-Internet"
    set srcintf "B2-SERVERS"
    set dstintf "port1"
    set srcaddr "B2-Servers"
    set dstaddr "all"
    set action accept
    set schedule "always"
    set service "DNS" "HTTP" "HTTPS" "NTP"
    set nat enable
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
   next
   edit 4
    set name "LAN-to-VPN"
    set srcintf "B2-USERS" "B2-SERVERS"
    set dstintf "Br2-to-HQ"
    set srcaddr "B2-Users" "B2-Servers"
    set dstaddr "HQ-LAN" "Branch1-LAN"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
    set logtraffic all
   next
   edit 5
    set name "VPN-to-LAN"
    set srcintf "Br2-to-HQ"
    set dstintf "B2-USERS" "B2-SERVERS"
    set srcaddr "HQ-LAN" "Branch1-LAN"
    set dstaddr "B2-Users" "B2-Servers"
    set action accept
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set ssl-ssh-profile "certificate-inspection"
    set av-profile "LAB-AV"
    set webfilter-profile "LAB-WEB"
    set ips-sensor "default"
    set application-list "LAB-APP"
```

```
    set logtraffic all
  next
  edit 6
    set name "Deny-All-Log"
    set srcintf "any"
    set dstintf "any"
    set srcaddr "all"
    set dstaddr "all"
    set action deny
    set schedule "always"
    set service "ALL"
    set logtraffic all
  next
end
```

Part 5: Verification after pasting

Cisco (all routers):

```
show ip interface brief
show ip route static
```

Every used interface must be `up/up`, not `administratively down`.

FortiGates:

```
get router info routing-table all
get vpn ipsec tunnel summary
get system ha status
show firewall policy
```

- The default route must be via port1 on every unit, with no default route via port5.
- Both tunnels must show `up` after the first ping across them.

Notes for your report:

- State that the tunnels use DES/MD5 because the lab VM did not accept AES, and that production would use AES256/SHA256.
- `123456` is only a lab key. Use a strong PSK and a strong `remote1` password.
- If SSL-VPN is not available on your FortiOS version, check `get system status` and use an IPsec dial-up tunnel with FortiClient instead.